

HireIQ

AI Hiring Intelligence Platform

Feature Ticket List — Sprint Backlog v1.0

Document Type	Feature Ticket List — Sprint Backlog
Product	HireIQ — AI Hiring Intelligence Platform
Version	v1.0.0
Total Tickets	60 tickets across 6 sprints
Total Story Points	252 points
Sprint Duration	2 weeks per sprint
Total Duration	13 weeks (Phases 1–6)
Date	June 14, 2026

TICKET LEGEND

Priority Levels, Types & Story Point Guide

Ticket Legend & Conventions

Priority Levels

P0 — Critical	Must be completed before anything else. Platform cannot function without this. Blocks all other work.
P1 — High	Core product feature. Must be in v1.0 launch. Sprint cannot close without this ticket done.
P2 — Medium	Important feature. Should be in v1.0. Can slip to next sprint if absolutely necessary.
P3 — Low	Nice-to-have for v1.0. Will move to v1.1 if sprint is over capacity.

Ticket Types

Feature	New user-facing functionality. Requires frontend + backend work.
Chore	Technical infrastructure, configuration, setup tasks. No direct user-facing impact.
Security	Security-focused implementation. Never de-prioritised. All Security tickets are P0 or P1.
AI	AI/ML specific implementation tickets. Covers model integration, scoring, NLP.
Infra	Infrastructure, DevOps, CI/CD, deployment, monitoring setup.
Bug	Defect found during development. Added to sprint in progress.

Story Point Scale (Modified Fibonacci)

1 point	Trivial change. Config update, text copy change, minor style fix. < 1 hour.
2 points	Simple task. Straightforward implementation, well-understood. 1–3 hours.
3 points	Standard task. Some complexity, may require research or multiple files. 3–6 hours.
5 points	Medium complexity. Involves multiple layers (route + service + schema + tests). 1–2 days.
8 points	High complexity. Multiple integrations, edge cases, thorough testing required. 2–3 days.
13 points	Very high complexity. Major feature with many moving parts. 3–5 days. Consider splitting.

Assignee Roles

BE	Backend Engineer — FastAPI, PostgreSQL, Redis, AWS S3 work
FE	Frontend Engineer — Next.js, React, TypeScript, Tailwind CSS work

ML	ML Engineer — AI models, NLP pipeline, scoring, recommendation system
DevOps	Infrastructure, CI/CD, Docker, Terraform, AWS configuration
Full	Full-stack — ticket touches both frontend and backend
Security	Security Engineer — OWASP, auth, encryption, penetration testing

MASTER BACKLOG

All 60 Tickets at a Glance

Master Backlog — All Tickets Summary

The following table lists all 60 tickets in the HireIQ v1.0 sprint backlog. Detailed ticket cards with acceptance criteria follow in subsequent sections.

ID	Title	Type	Priority	Points	Sprint	Assignee
HIQ-001	Project scaffolding — monorepo setup + Docker Compose	Chore	P0	3	S1	DevOps
HIQ-002	PostgreSQL schema — run all migrations via Alembic	Chore	P0	3	S1	BE
HIQ-003	FastAPI app factory, router registration, health endpoint	Chore	P0	3	S1	BE
HIQ-004	Redis connection pool setup + helper utilities	Chore	P0	2	S1	BE
HIQ-005	Pydantic BaseSettings — environment variable configuration	Chore	P0	2	S1	BE
HIQ-006	Structlog JSON logging middleware + request ID injection	Chore	P1	3	S1	BE
HIQ-007	GitHub Actions CI pipeline — lint + test + build on PR	Infra	P0	5	S1	DevOps
HIQ-008	Next.js 14 project setup — TypeScript, Tailwind, ESLint	Chore	P0	3	S1	FE
HIQ-009	Design system — Tailwind config, tokens, dark mode toggle	Feature	P1	3	S1	FE
HIQ-010	Zustand stores — authStore + uiStore setup	Chore	P0	2	S1	FE
HIQ-011	POST /auth/register — user registration with bcrypt	Feature	P0	5	S2	BE
HIQ-012	POST /auth/login — JWT + httpOnly refresh cookie	Feature	P0	5	S2	BE
HIQ-013	POST /auth/refresh — token rotation via Redis	Feature	P0	3	S2	BE
HIQ-014	POST /auth/logout — Redis token invalidation	Feature	P0	2	S2	BE
HIQ-015	JWT auth middleware — get_current_user dependency	Security	P0	3	S2	BE
HIQ-016	RBAC middleware — require_role decorator	Security	P0	3	S2	BE

HIQ-017	Rate limiting middleware — Redis sliding window	Security	P1	5	S2	BE
HIQ-018	Global error handler — ValidationError + 500 envelope	Chore	P0	3	S2	BE
HIQ-019	Login page UI — split screen, form, validation	Feature	P0	5	S2	FE
HIQ-020	Register page UI — role selection + step 2 form	Feature	P0	5	S2	FE
HIQ-021	GET + PUT /users/profile — profile CRUD	Feature	P0	5	S3	BE
HIQ-022	POST /users/avatar — S3 avatar upload	Feature	P2	3	S3	BE
HIQ-023	Profile completion % calculator — service + update hook	Feature	P1	3	S3	BE
HIQ-024	Job model + Alembic migration	Chore	P0	2	S3	BE
HIQ-025	POST /jobs — create job (recruiter only)	Feature	P0	5	S3	BE
HIQ-026	GET /jobs — paginated list with search + filters	Feature	P0	8	S3	BE
HIQ-027	GET /jobs/:id — job detail endpoint	Feature	P0	3	S3	BE
HIQ-028	PUT /jobs/:id + DELETE /jobs/:id — owner-only update/delete	Feature	P0	5	S3	BE
HIQ-029	PATCH /jobs/:id/close — close job endpoint	Feature	P1	2	S3	BE
HIQ-030	Job listing page UI — cards, search bar, filter sidebar	Feature	P0	8	S3	FE
HIQ-031	AWS S3 bucket setup + IAM role + bucket policy	Infra	P0	3	S4	DevOps
HIQ-032	POST /resumes/upload — S3 pre-signed URL + file validation	Feature	P0	8	S4	BE
HIQ-033	GET + DELETE /resumes/me — resume retrieval + deletion	Feature	P0	3	S4	BE
HIQ-034	PyMuPDF text extraction utility + unit tests	Chore	P0	3	S4	BE
HIQ-035	Resume async AI parsing trigger (BackgroundTasks)	Feature	P1	5	S4	BE
HIQ-036	Application model + migration + UNIQUE constraint	Chore	P0	2	S4	BE
HIQ-037	POST /applications — apply with all 6 validation checks	Feature	P0	8	S4	BE
HIQ-038	GET /applications/me — job seeker application history	Feature	P0	3	S4	BE
HIQ-039	GET /applications/job/:id — recruiter applicant list	Feature	P0	5	S4	BE

HIQ-040	PATCH /applications/:id/status — status update	Feature	P0	3	S4	BE
HIQ-041	AI Service: FastAPI app scaffold + model loader	Chore	P0	5	S5	ML
HIQ-042	Resume Intelligence Engine: spaCy NER + skill extraction	AI	P0	13	S5	ML
HIQ-043	Candidate Scoring Engine: TF-IDF + cosine similarity	AI	P0	13	S5	ML
HIQ-044	Job Recommendation Engine: content-based filtering	AI	P1	8	S5	ML
HIQ-045	AI Interview Evaluator: sentence-transformer scoring	AI	P1	8	S5	ML
HIQ-046	POST /ai/analyze-resume endpoint + schema	Feature	P0	3	S5	ML
HIQ-047	POST /ai/score-candidate endpoint + Redis cache	Feature	P0	3	S5	ML
HIQ-048	POST /ai/recommend-jobs endpoint	Feature	P1	3	S5	ML
HIQ-049	POST /ai/evaluate-interview endpoint	Feature	P1	3	S5	ML
HIQ-050	AI Service client (httpx) in Core API + fallback logic	Feature	P0	5	S5	BE
HIQ-051	Job seeker dashboard UI — stats, AI score ring, recs	Feature	P0	8	S6	FE
HIQ-052	Recruiter dashboard UI — jobs table, charts, top candidates	Feature	P0	8	S6	FE
HIQ-053	Resume upload page UI — drag & drop + progress + AI results	Feature	P0	8	S6	FE
HIQ-054	My Applications page UI — table, status tabs, badges	Feature	P0	5	S6	FE
HIQ-055	AI Insights page UI — score gauge, skill gap, roadmap	Feature	P1	8	S6	FE
HIQ-056	AI Interview chat UI — Q&A bubbles, progress, results	Feature	P1	8	S6	FE
HIQ-057	Recruiter Applicants page — ranked table, drawer, bulk actions	Feature	P0	8	S6	FE
HIQ-058	Recruiter Analytics page — 5 Recharts components	Feature	P1	5	S6	FE
HIQ-059	AWS ECS + RDS + ElastiCache + CloudFront production setup	Infra	P0	8	S6	DevOps
HIQ-060	OWASP Top 10 audit + security headers + pentest checklist	Security	P0	8	S6	Security

SPRINT 1

Foundation & Project Setup | Weeks 1–2 | Goal: Running skeleton on localhost

Sprint 1 — Foundation & Project Setup

Sprint 1 Weeks 1–2

Sprint Goal: Have a running multi-service skeleton (API + Frontend + DB + Redis) with CI passing on every commit.

HIQ-001 Project Scaffolding — Monorepo + Docker Compose

Type: Chore

Priority: P0

Points: 3

Sprint: Sprint 1 | Assignee: DevOps

Description

Set up the monorepo structure with three root directories (backend/, ai_service/, frontend/) and a root docker-compose.yml that starts all services including PostgreSQL and Redis with a single docker-compose up command.

Acceptance Criteria

1. docker-compose up starts all 5 services (api, ai, frontend, db, redis) without errors
2. All services can communicate with each other on the Docker network
3. Each service has its own Dockerfile with multi-stage build (deps + app layers)
4. README.md updated with local setup instructions (clone, cp .env.example .env, docker-compose up)
5. .gitignore covers .env files, __pycache__, node_modules, .next

Dependencies

None

HIQ-002 PostgreSQL Schema — Alembic Migrations

Type: Chore

Priority: P0

Points: 3

Sprint: Sprint 1 | Assignee: BE

Description

Configure Alembic for database migration management and create the initial migration file containing all five tables: users, jobs, resumes, applications, interview_sessions — with all indexes as defined in the BSD.

Acceptance Criteria

6. alembic upgrade head runs successfully against a fresh PostgreSQL database
7. All 5 tables created with correct column types, constraints, and defaults
8. All indexes created: GIN for skills and full-text, B-tree for FK lookups and sorting
9. UNIQUE constraint on applications(user_id, job_id) confirmed via psql
10. alembic downgrade -1 cleanly reverses the migration

Dependencies

- [HIQ-001 — Docker Compose with PostgreSQL running](#)

HIQ-003 FastAPI App Factory + Router Registration + Health Endpoint

Type: Chore	Priority: P0	Points: 3	Sprint: Sprint 1 Assignee: BE
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Description

Implement the FastAPI application factory in main.py with CORS middleware, router registration for all 7 route groups, and a GET /health endpoint. Lifespan context manager handles startup/shutdown of DB and Redis connections.

Acceptance Criteria

11. GET /health returns HTTP 200 with { status: ok, version: 1.0.0 }
12. All 7 routers registered at correct prefixes (/v1/auth, /v1/jobs, etc.)
13. CORS allows localhost:3000 and rejects all other origins
14. Swagger docs accessible at /docs (dev only)
15. App starts without errors via uvicorn main:app --reload

Dependencies

- [HIQ-001](#)
- [HIQ-002](#)
- [HIQ-004](#)
- [HIQ-005](#)

HIQ-007 GitHub Actions CI Pipeline

Type: Infra	Priority: P0	Points: 5	Sprint: Sprint 1 Assignee: DevOps
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Description

Create a GitHub Actions workflow file (.github/workflows/ci.yml) that runs on every push to main and on all PRs. The pipeline must run: Python linting (ruff), TypeScript linting (eslint), backend unit tests (pytest), frontend unit tests (jest), and Docker build validation.

Acceptance Criteria

16. Pipeline triggers on push to main and pull_request to main
17. ruff check backend/ passes with zero violations
18. eslint frontend/ passes with zero violations
19. pytest tests/ with --cov=app --cov-fail-under=70 passes
20. jest --coverage passes with 70% threshold
21. docker build ./backend and docker build ./frontend succeed
22. Pipeline fails fast: if lint fails, tests do not run
23. Pipeline runtime under 5 minutes on GitHub-hosted runners

Dependencies

- [HIQ-001](#)
- [HIQ-003](#)
- [HIQ-008](#)

HIQ-008 Next.js 14 Project Setup — TypeScript + Tailwind + ESLint

Type: Chore

Priority: P0

Points: 3

Sprint: Sprint 1 | Assignee: FE

Description

Bootstrap the Next.js 14 application with App Router, TypeScript in strict mode, Tailwind CSS v3 with custom HireIQ design tokens, ESLint with Next.js config, and all required dependencies installed and pinned in package.json.

Acceptance Criteria

- 24. npx next dev starts the app on port 3000 without TypeScript errors
- 25. tailwind.config.ts contains all HireIQ brand color tokens from FSD Section 1
- 26. TypeScript strict mode enabled in tsconfig.json
- 27. All dependencies pinned to specific versions in package.json
- 28. shadcn/ui initialised with default theme
- 29. Folder structure matches FSD Section 6.3 exactly

Dependencies

- [HIQ-001](#)

SPRINT 2

Authentication & Security | Weeks 3–4 | Goal: Secure JWT auth with RBAC

Sprint 2 — Authentication & Security

Sprint 2 Weeks 3–4

Sprint Goal: Complete JWT authentication system with registration, login, refresh, RBAC, rate limiting, and passing auth test suite.

HIQ-011 POST /auth/register — User Registration

Type: Feature

Priority: P0

Points: 5

Sprint: Sprint 2 | Assignee: BE

Description

Implement user registration endpoint accepting name, email, password, and role. Hash password with bcrypt cost=12. Check for duplicate email. Issue JWT access token and set refresh token as httpOnly cookie. Store refresh token hash in Redis.

Acceptance Criteria

- 30. POST /auth/register with valid data returns HTTP 201 with { access_token, user }
- 31. refresh_token set as httpOnly, secure, SameSite=Lax cookie in response
- 32. Password stored as bcrypt hash (cost=12) — never plain text
- 33. Duplicate email registration returns HTTP 409 with code EMAIL_EXISTS
- 34. Missing required fields return HTTP 422 with field-level error details
- 35. Invalid role value returns HTTP 422
- 36. Unit test coverage: minimum 95% on auth routes

Dependencies

- [HIQ-002](#)
- [HIQ-003](#)
- [HIQ-004](#)
- [HIQ-005](#)
- [HIQ-018](#)

HIQ-012 POST /auth/login — JWT Authentication

Type: Feature

Priority: P0

Points: 5

Sprint: Sprint 2 | Assignee: BE

Description

Implement login endpoint with timing attack prevention (always run bcrypt.verify even if user not found), failed attempt tracking in Redis, and account lockout after 5 consecutive failures. Issue new token pair on success.

Acceptance Criteria

- 37. POST /auth/login with correct credentials returns HTTP 200 with access_token

- 38. Wrong password returns HTTP 401 with generic message (no hint about which field failed)
- 39. Response time for wrong password is equal to correct password (timing attack prevention)
- 40. After 5 failed attempts for same email: account locked for 15 minutes, returns 429
- 41. Locked account returns 429 even with correct password during lockout period
- 42. Successful login resets the failed attempt counter in Redis

Dependencies

- [HIQ-011](#)
- [HIQ-015](#)
- [HIQ-017](#)

HIQ-015 JWT Auth Middleware — get_current_user Dependency

Type: Security

Priority: P0

Points: 3

Sprint: Sprint 2 | Assignee: BE

Description

Implement the get_current_user FastAPI dependency that extracts the Bearer token from the Authorization header, decodes and validates it using python-jose, and returns the authenticated User ORM object. Used as Depends() on all protected routes.

Acceptance Criteria

- 43. Valid JWT returns the User object with correct id and role
- 44. Missing Authorization header returns HTTP 401 with code INVALID_TOKEN
- 45. Expired JWT returns HTTP 401 with code INVALID_TOKEN
- 46. Tampered JWT signature (manually modified) returns HTTP 401
- 47. Valid JWT for deactivated user (is_active=False) returns HTTP 401
- 48. Middleware adds < 5ms overhead to request processing time

Dependencies

- [HIQ-011](#)
- [HIQ-012](#)
- [HIQ-005](#)

HIQ-016 RBAC Middleware — require_role Decorator

Type: Security

Priority: P0

Points: 3

Sprint: Sprint 2 | Assignee: BE

Description

Implement the require_role decorator that wraps FastAPI route handlers and checks the current user's role against an allowed roles list. Also implement verify_resource_owner helper for ownership checks.

Acceptance Criteria

- 49. Route decorated with @require_role('recruiter') returns 403 when called by job_seeker
- 50. Route decorated with @require_role('recruiter','admin') allows both roles
- 51. Error response includes required roles and actual role for debugging
- 52. verify_resource_owner raises 403 when user.id != resource owner_id (and user is not admin)
- 53. Admin role bypasses ownership checks in verify_resource_owner
- 54. Tests cover all role combinations for at least 3 different protected endpoints

Dependencies

- [HIQ-015](#)

HIQ-019 Login Page UI

Type: Feature

Priority: P0

Points: 5

Sprint: Sprint 2 | Assignee: FE

Description

Build the login page at /auth/login with a split-screen layout. Left panel: blue gradient hero with HireIQ branding. Right panel: login form with email/password fields, show/hide password toggle, Remember Me checkbox, loading state on submit, and inline validation errors.

Acceptance Criteria

- 55. Page renders correctly on desktop (1280px+) and mobile (375px)
- 56. Email field: required validation + email format check on blur
- 57. Password field: required validation + show/hide toggle works
- 58. Submit button shows spinner and disables on form submission
- 59. API error (401) displays in red alert box below form
- 60. Successful login redirects to /dashboard
- 61. returnUrl query param redirect works correctly
- 62. Dark mode: all elements render correctly with dark: classes

Dependencies

- [HIQ-010](#)
- [HIQ-008](#)
- [HIQ-009](#)

HIQ-020 Register Page UI — Role Selection + Form

Type: Feature

Priority: P0

Points: 5

Sprint: Sprint 2 | Assignee: FE

Description

Build the registration page at /auth/register with a 2-step flow: Step 1 shows role selection cards (Job Seeker / Recruiter), Step 2 shows the registration form with name, email, password with strength meter, confirm password, and 3 consent checkboxes.

Acceptance Criteria

- 63. Step 1: Two large role cards displayed side by side on desktop, stacked on mobile
- 64. Selected role card has blue border + checkmark icon indicator
- 65. Step 2 only visible after role selection
- 66. Password strength meter updates in real time using zxcvbn (4 levels)
- 67. All 3 consent checkboxes must be checked before form submits
- 68. Confirm password mismatch shows inline error
- 69. Successful registration shows success animation then redirects to /dashboard
- 70. API duplicate email error shows inline alert

Dependencies

- [HIQ-010](#)
- [HIQ-008](#)
- [HIQ-009](#)

SPRINT 3

User Profiles & Jobs Module | Weeks 5–6 | Goal: Full job CRUD with search

Sprint 3 — User Profiles & Jobs Module

Sprint 3 Weeks 5–6

Sprint Goal: Complete user profile management and fully functional job CRUD with search, filter, and pagination working end-to-end.

HIQ-021 GET + PUT /users/profile — Profile CRUD

Type: Feature

Priority: P0

Points: 5

Sprint: Sprint 3 | Assignee: BE

Description

Implement profile retrieval and update endpoints. Profile update accepts partial fields (name, bio, skills, experience, education). On every update, recalculate and persist the profile_completion_percentage.

Acceptance Criteria

- 71. GET /users/profile returns full user object excluding password_hash
- 72. PUT /users/profile with partial fields updates only those fields (others unchanged)
- 73. Skills array accepts up to 50 items, each max 50 characters
- 74. Profile completion % recalculated:
 $\text{name}(10) + \text{email}(10) + \text{bio}(15) + \text{skills}(20) + \text{experience}(20) + \text{education}(15) + \text{avatar}(10) = 100$
- 75. Updated profile invalidates the Redis cache key user_profile:{user_id}
- 76. Response excludes sensitive fields: password_hash, internal IDs

Dependencies

- HIQ-011
- HIQ-015
- HIQ-002

HIQ-026 GET /jobs — Paginated List with Search + Filters

Type: Feature

Priority: P0

Points: 8

Sprint: Sprint 3 | Assignee: BE

Description

Implement the job listing endpoint with full-text search across title and description using PostgreSQL tsvector, skills array containment filter, location partial match, salary range overlap filter, and multiple sort options. Results must be paginated and Redis-cached.

Acceptance Criteria

- 77. GET /jobs?page=1&limit=10 returns first 10 active jobs with pagination metadata
- 78. GET /jobs?search=python returns jobs with python in title or description
- 79. GET /jobs?skills=Python&skills=React returns jobs requiring BOTH skills
- 80. GET /jobs?salary_min=50000 returns jobs where salary_max >= 50000

- 81. GET /jobs?location=Mumbai returns case-insensitive partial location matches
- 82. GET /jobs?sort_by=salary returns jobs sorted by salary_max descending
- 83. Results cached in Redis for 5 minutes with filter-hash cache key
- 84. Cache invalidated when any job is created, updated, or deleted
- 85. Returns 200 with empty data array (not 404) when no jobs match filters
- 86. Pagination: page 1 and page 2 results are non-overlapping

Dependencies

- [HIQ-025](#)
- [HIQ-002](#)
- [HIQ-004](#)

HIQ-025 POST /jobs — Create Job Posting

Type: Feature

Priority: P0

Points: 5

Sprint: Sprint 3 | Assignee: BE

Description

Implement job creation endpoint restricted to recruiter and admin roles. Validates all required fields via Pydantic JobCreate schema. Stores job with recruiter_id from authenticated user. Invalidates job listing cache.

Acceptance Criteria

- 87. POST /jobs by recruiter returns HTTP 201 with created job object
- 88. POST /jobs by job_seeker returns HTTP 403 with INSUFFICIENT_ROLE code
- 89. title shorter than 3 characters returns HTTP 422
- 90. description shorter than 50 characters returns HTTP 422
- 91. required_skills empty array returns HTTP 422
- 92. salary_max less than salary_min returns HTTP 422
- 93. job_type with invalid value returns HTTP 422
- 94. Created job has status='active' by default
- 95. Job listing Redis cache invalidated after successful creation

Dependencies

- [HIQ-015](#)
- [HIQ-016](#)
- [HIQ-002](#)

HIQ-030 Job Listing Page UI — Cards, Search, Filters

Type: Feature

Priority: P0

Points: 8

Sprint: Sprint 3 | Assignee: FE

Description

Build the complete job listing page at /jobs. Includes a sticky search bar at top, filter sidebar on desktop (collapsible bottom sheet on mobile), job cards in a responsive grid, sort dropdown, skeleton loaders during fetch, empty state illustration, and pagination controls.

Acceptance Criteria

- 96. Search input debounces 300ms before triggering API call
- 97. Filter sidebar: skills multi-select, location text, salary range slider, job type checkboxes
- 98. Active filters shown as dismissable chips above job cards

99. Sort dropdown: Most Recent / Highest Salary / Most Applicants

100. JobCard renders correctly with all fields: title, company, salary, skills, date, applicants

101. Skeleton loaders (3 cards) shown while API call is in progress

102. Empty state: illustration + message + Clear Filters button when no results

103. Pagination: Previous / page numbers / Next buttons, URL updates with ?page=N

104. Mobile: filters in bottom sheet modal, single-column card grid

Dependencies

- [HIQ-008](#)
- [HIQ-009](#)
- [HIQ-010](#)
- [HIQ-026](#)

SPRINT 4

Resume & Applications Module | Weeks 7–8 | Goal: Upload resume, apply to jobs

Sprint 4 — Resume & Applications Module

Sprint 4 Weeks 7–8

Sprint Goal: Full resume upload pipeline (S3), PDF parsing, and complete job application flow with all validation checks.

HIQ-031 AWS S3 Bucket Setup + IAM Role + Bucket Policy

Type: Infra

Priority: P0

Points: 3

Sprint: Sprint 4 | Assignee: DevOps

Description

Provision the AWS S3 bucket (hireiq-resumes-prod), configure the bucket policy to deny all public access, enable server-side encryption (SSE-S3), enable versioning, and create an IAM role for the ECS task with minimal S3 permissions.

Acceptance Criteria

105. S3 bucket created in ap-south-1 region with versioning enabled
106. Public access block enabled on bucket (all 4 settings = true)
107. Bucket policy: only the hireiq-api IAM role can GetObject and PutObject
108. SSE-S3 encryption enforced — uploads without encryption rejected by bucket policy
109. IAM role created with minimum permissions: s3:PutObject, s3:GetObject, s3:DeleteObject on hireiq-resumes-prod/* only
110. Terraform or CloudFormation template committed to /infrastructure directory

Dependencies

- [HIQ-001](#)
- [HIQ-007](#)

HIQ-032 POST /resumes/upload — S3 Upload + Validation

Type: Feature

Priority: P0

Points: 8

Sprint: Sprint 4 | Assignee: BE

Description

Implement the resume upload endpoint. Validate file type (PDF only) and size (max 5MB) before any processing. Generate an S3 pre-signed PUT URL. Save file URL to database. Trigger async AI parsing via FastAPI BackgroundTasks.

Acceptance Criteria

111. POST /resumes/upload with valid PDF returns HTTP 201 with { resume_id, file_url, parse_status: pending }
112. Upload of non-PDF file (any other content-type) returns HTTP 400 with INVALID_FILE_TYPE

- 113. Upload of file > 5MB returns HTTP 400 with FILE_TOO_LARGE and actual size in message
- 114. File URL stored in resumes table with parse_status='pending'
- 115. If user already has a resume: old S3 file deleted in background, record updated
- 116. Async AI parsing triggered in BackgroundTasks (does not block the HTTP response)
- 117. Resume model user_id UNIQUE constraint: each user has max one active resume

Dependencies

- [HIQ-031](#)
- [HIQ-034](#)
- [HIQ-035](#)
- [HIQ-015](#)

HIQ-037 POST /applications — Apply with 6 Validation Checks

Type: Feature

Priority: P0

Points: 8

Sprint: Sprint 4 | Assignee: BE

Description

Implement the job application endpoint with all 6 pre-flight validation checks in order: job exists, job is active, user has resume, no duplicate application. Create application record. Increment job applicant count. Trigger async AI scoring.

Acceptance Criteria

- 118. POST /applications with job_id returns HTTP 201 with { application_id, status: applied }
- 119. Apply to non-existent job_id returns HTTP 404
- 120. Apply to closed job returns HTTP 400 with JOB_CLOSED code
- 121. Apply without uploaded resume returns HTTP 400 with NO_RESUME code
- 122. Apply to same job twice returns HTTP 409 with DUPLICATE_APPLICATION code
- 123. Job applicant_count incremented by 1 after successful application
- 124. AI scoring triggered asynchronously — does not block the 201 response
- 125. application.ai_score updated in DB within 10 seconds of submission

Dependencies

- [HIQ-032](#)
- [HIQ-036](#)
- [HIQ-025](#)
- [HIQ-015](#)
- [HIQ-016](#)
- [HIQ-050](#)

HIQ-053 Resume Upload Page UI — Drag & Drop + AI Results

Type: Feature

Priority: P0

Points: 8

Sprint: Sprint 4 | Assignee: FE

Description

Build the resume upload page at /resume with a drag-and-drop upload zone (react-dropzone), real-time upload progress bar, AI analysis results display (parsed skills, experience, education, score ring), and improvement suggestions section.

Acceptance Criteria

- 126. Drag & drop zone accepts PDF files only — shows error for other types
- 127. File size > 5MB blocked client-side with error message before API call
- 128. Upload progress bar shows 0–100% during upload
- 129. After upload: AI Analyzing... spinner shown while parse_status = pending
- 130. After parsing complete: skills tags, experience card, education card displayed
- 131. AI Score ring animated from 0 to score value on mount
- 132. If score < 70: improvement suggestions section visible with bulleted tips
- 133. Replace Resume flow: confirmation dialog before replacing existing resume

Dependencies

- [HIQ-032](#)
- [HIQ-033](#)
- [HIQ-008](#)
- [HIQ-009](#)
- [HIQ-010](#)

SPRINT 5

AI / ML System | Weeks 9–10 | Goal: All 4 AI engines live and tested

Sprint 5 — AI / ML System

Sprint 5 Weeks 9–10

Sprint Goal: All four AI engines implemented and integrated: resume parsing, candidate scoring, job recommendations, and interview evaluation.

HIQ-042 Resume Intelligence Engine — spaCy NER + Skill Extraction

Type: AI	Priority: P0	Points: 13	Sprint: Sprint 5 Assignee: ML
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Description

Build the resume intelligence engine that converts raw PDF text into structured JSON. Uses spaCy `en_core_web_lg` for Named Entity Recognition and a curated 2,000+ skill dictionary for skill extraction. Handles experience and education detection via regex and NER.

Acceptance Criteria

134. Input: raw resume text string. Output: { skills[], experience, education, projects[] }
135. Skills dictionary covers at least 2,000 technology and business skills
136. Skill extraction accuracy $\geq 85\%$ on a 50-resume test set (verified by ML engineer)
137. Experience extraction handles: date ranges (2019-2023), text patterns (3 years of experience)
138. Education extraction identifies: B.Tech, M.Sc, MBA, PhD and equivalent patterns
139. Projects section extracted when header keywords (Projects, Portfolio) are detected
140. Engine processes a 3-page resume in under 2 seconds
141. Unit tests cover: tech resume, management resume, fresh graduate resume, non-English partial text

Dependencies

- [HIQ-041](#)
- [HIQ-034](#)

HIQ-043 Candidate Scoring Engine — TF-IDF + Cosine Similarity

Type: AI	Priority: P0	Points: 13	Sprint: Sprint 5 Assignee: ML
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Description

Build the candidate scoring engine that computes a 0–100 match score between a resume and a job description using TF-IDF vectorization and cosine similarity. Also produces `matched_skills` and `missing_skills` arrays.

Acceptance Criteria

- 142. Input: { resume_text, job_description }. Output: { score: int, matched_skills[], missing_skills[] }
- 143. Score is always an integer between 0 and 100 (never null, never float)
- 144. TF-IDF model pre-fitted on a corpus of at least 5,000 job descriptions
- 145. Score correlation ≥ 0.80 with human expert scores on a 30-pair test set
- 146. matched_skills = intersection of resume skills and job required_skills
- 147. missing_skills = job required_skills not found in resume skills
- 148. Inference time $< 500\text{ms}$ per request (model loaded at startup, not per request)
- 149. Model serialized as tfidf_vectorizer.pkl using joblib for persistence

Dependencies

- [HIQ-041](#)
- [HIQ-042](#)

HIQ-044 Job Recommendation Engine — Content-Based Filtering

Type: AI	Priority: P1	Points: 8	Sprint: Sprint 5 Assignee: ML
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Description

Build the job recommendation engine that returns the top 5 most relevant jobs for a given user based on their skill profile. Uses content-based filtering: vectorize user skills, compute similarity against all active job vectors stored in Redis.

Acceptance Criteria

- 150. Input: { user_id, skills[] }. Output: [{ job_id, title, match_score }] (top 5)
- 151. Uses the same TF-IDF vectorizer as scoring engine for consistency
- 152. Job matrix cached in Redis and updated when new jobs are created
- 153. Returns at most 5 recommendations, ordered by similarity score descending
- 154. If user has fewer than 3 skills: returns most recent 5 active jobs as fallback
- 155. Cache invalidated for a user when their skills profile is updated
- 156. Unit test: user with Python+React skills gets Python/React jobs ranked above Java jobs

Dependencies

- [HIQ-043](#)
- [HIQ-041](#)

HIQ-045 AI Interview Evaluator — Sentence Transformer Scoring

Type: AI	Priority: P1	Points: 8	Sprint: Sprint 5 Assignee: ML
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Description

Build the interview evaluation engine using Sentence-Transformers (all-MiniLM-L6-v2) to encode candidate answers and compare them against a curated ideal answer corpus. Returns score, feedback, and confidence label.

Acceptance Criteria

- 157. Input: { question, answer }. Output: { score: int, feedback: string, confidence: low|medium|high }
- 158. Sentence-Transformers model loaded at service startup (not per request)
- 159. Ideal answer corpus covers at least 50 common interview questions across 5 domains

- 160. Score formula: (semantic_similarity * 85) + (keyword_coverage * 15), clamped 0-100
- 161. Feedback rules: >= 80 Excellent, 60-79 Good but lacks depth, 40-59 Needs more detail, < 40 revisit fundamentals
- 162. Confidence: high if score >= 75, medium if 50-74, low if < 50
- 163. Inference time < 2 seconds per question-answer pair
- 164. Unit tests cover: excellent answer, mediocre answer, completely wrong answer, empty answer

Dependencies

- [HIQ-041](#)

HIQ-050 AI Service HTTP Client in Core API

Type: Feature

Priority: P0

Points: 5

Sprint: Sprint 5 | Assignee: BE

Description

Implement the AIService class in the Core API using httpx async client to call all 4 AI service endpoints. Implement graceful fallback for timeouts and errors (AI failure must not fail the user-facing operation). Integrate Redis caching for scores and recommendations.

Acceptance Criteria

- 165. analyze_resume: calls AI service, returns structured dict, returns empty dict on failure (does not raise)
- 166. score_candidate: checks Redis cache first, calls AI service on miss, caches result 1hr
- 167. recommend_jobs: checks Redis cache first, calls AI service on miss, caches result 30min
- 168. evaluate_interview: calls AI service, raises HTTPException only if AI service is completely unavailable
- 169. All calls use httpx.AsyncClient with 10-second timeout
- 170. Timeout and HTTP errors logged at ERROR level with request context
- 171. Integration test: mock AI service with httpx.MockTransport, verify fallback returns

Dependencies

- [HIQ-041](#)
- [HIQ-046](#)
- [HIQ-047](#)
- [HIQ-048](#)
- [HIQ-049](#)
- [HIQ-004](#)

SPRINT 6

Frontend Dashboards, AI UI & Production Deploy | Weeks 11–13

Sprint 6 — Dashboards, AI UI & Production Deployment

Sprint 6 Weeks 11–13

Sprint Goal: Complete all dashboard UIs, AI insight pages, interview chat interface, and deploy the full platform to AWS production.

HIQ-051 Job Seeker Dashboard UI

Type: Feature

Priority: P0

Points: 8

Sprint: Sprint 6 | Assignee: FE

Description

Build the complete job seeker dashboard at /dashboard. Includes animated stats cards, AI score ring with animated draw, profile completion banner, recommended jobs section (3 cards), recent applications table, skill gap alert box, resume status card, and quick action buttons.

Acceptance Criteria

- 172. All 4 stats cards render with animated count-up on mount
- 173. AI score ring draws from 0 to score value over 1.5 seconds on page load
- 174. Score ring color: green ≥ 70 , orange 40-69, red < 40
- 175. Profile completion banner visible and dismissable if completion $< 80\%$
- 176. Recommended jobs: 3 JobCard components with AI match % badge
- 177. Recent applications table shows last 5 with status color badges
- 178. Skill gap alert: orange info box with top 3 missing skills if any exist
- 179. All sections show skeleton loaders while data is fetching
- 180. All sections show empty states if no data is available

Dependencies

- [HIQ-008](#)
- [HIQ-009](#)
- [HIQ-010](#)
- [HIQ-021](#)
- [HIQ-038](#)
- [HIQ-050](#)

HIQ-052 Recruiter Dashboard UI

Type: Feature

Priority: P0

Points: 8

Sprint: Sprint 6 | Assignee: FE

Description

Build the complete recruiter dashboard at /dashboard. Includes stats cards, active jobs table with sort and quick-filter, top candidates widget, applicant trend line chart, score distribution bar chart, and skill tag cloud.

Acceptance Criteria

- 181. 4 KPI cards: Active Jobs, Total Applications, Shortlisted, Average AI Score
- 182. Active jobs table: sortable by column headers, filter tabs (All/Active/Closed)
- 183. Applicant Trend chart: Recharts LineChart, last 30 days, daily count, hover tooltip
- 184. Score Distribution chart: Recharts BarChart, 5 score buckets, color-coded bars
- 185. Top 5 candidates widget: CandidateCard components, click opens drawer
- 186. Skill tag cloud: top 10 skills with size proportional to frequency
- 187. Post New Job primary CTA button in top-right, links to /jobs/create
- 188. All charts fully responsive: resize correctly at all breakpoints

Dependencies

- [HIQ-008](#)
- [HIQ-009](#)
- [HIQ-010](#)
- [HIQ-039](#)
- [HIQ-050](#)

HIQ-055 AI Insights Page UI

Type: Feature	Priority: P1	Points: 8	Sprint: Sprint 6 Assignee: FE
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Description

Build the AI Insights page at /ai/insights — the showcase page of the platform. Large animated score gauge, 4 sub-score cards, two-column skill gap analysis, job match breakdown bar chart, improvement roadmap numbered list, industry benchmark comparison, and PDF export button.

Acceptance Criteria

- 189. Score gauge: large SVG circle (160px), animates on mount, color by score band
- 190. 4 sub-score cards: Skills Match %, Experience Score, Education Score, Profile Completeness
- 191. Skill gap: left column green skill tags (have), right column orange tags (missing)
- 192. Missing skills link to external learning resources (Coursera/Udemy)
- 193. Job match breakdown: horizontal Recharts BarChart by category
- 194. Improvement roadmap: numbered list, each item has a completion checkbox (saved to localStorage)
- 195. Industry benchmark: bar chart comparing user vs average vs top 10%
- 196. Re-analyze button triggers new AI analysis, shows spinner, updates all scores
- 197. Export PDF button generates downloadable report

Dependencies

- [HIQ-051](#)
- [HIQ-050](#)
- [HIQ-042](#)
- [HIQ-043](#)

HIQ-056 AI Interview Chat UI

Type: Feature	Priority: P1	Points: 8	Sprint: Sprint 6 Assignee: FE
Description Build the AI mock interview page at /ai/interview with a full-height chat interface. Setup panel for interview type/difficulty/questions, chat window with AI question and user answer bubbles, progress bar, scoring feedback after each answer, and session completion summary.			
Acceptance Criteria - Setup panel: dropdown for interview type, difficulty, and question count - Start Interview button begins session and hides setup panel - AI question bubbles: left-aligned, bg-slate-100, show question number - User answer input: large textarea, send on Ctrl+Enter or Send button, 1000 char limit - After answer submit: AI feedback bubble appears with score badge + feedback text + confidence - Progress bar shows X of Y questions complete - Session timer shows elapsed time - After final question: completion screen with overall score + per-question breakdown table - Interview history in left sidebar: past sessions with date, type, score badge			
Dependencies - HIQ-051 - HIQ-050 - HIQ-045			

HIQ-057 Recruiter Applicants Page — Ranked Table + Drawer			
Type: Feature	Priority: P0	Points: 8	Sprint: Sprint 6 Assignee: FE
Description Build the recruiter applicants page at /recruiter/applicants/:jobId. Shows all applicants ranked by AI score with color-coded score badges, matched/missing skills, one-click shortlist/reject buttons with optimistic UI, candidate detail drawer, bulk actions, and CSV export.			
Acceptance Criteria - Candidates sorted by ai_score descending by default - Score badge: circle colored green/blue/orange/red by score band - Matched skills shown as green tags (max 5 + count), missing as orange tags - Shortlist button: optimistic UI updates status instantly, API call in background - Undo toast appears for 5 seconds after status change with Undo button - Click candidate name: right-side drawer slides in with full profile + resume link - Row checkboxes: select multiple, Shortlist Selected / Reject Selected bulk buttons appear - Bulk actions: confirmation dialog before executing, progress indicator during - Export to CSV: downloads file with all visible columns			
Dependencies - HIQ-052 - HIQ-039 - HIQ-040 - HIQ-050			

HIQ-059 AWS Production Infrastructure Setup

Type: Infra

Priority: P0

Points: 8

Sprint: Sprint 6 | Assignee: DevOps

Description

Provision the complete AWS production infrastructure using Terraform: VPC with public/private subnets, Application Load Balancer, ECS Fargate cluster with Core API and AI Service task definitions, RDS PostgreSQL Multi-AZ, ElastiCache Redis, S3 (already done in HIQ-031), CloudFront, and Route 53.

Acceptance Criteria

- 216. Terraform plan shows no errors, terraform apply completes without manual steps
- 217. All services deployed in private subnets — no direct public internet access
- 218. ALB routes /v1/* to Core API ECS, /ai/* to AI ECS service
- 219. RDS Multi-AZ enabled, automated daily snapshots configured with 30-day retention
- 220. ElastiCache Redis single-node in private subnet, in-transit encryption enabled
- 221. CloudFront distribution serving frontend with cache TTL 1hr for static assets
- 222. Route 53 records: hireiq.io to CloudFront, api.hireiq.io to ALB
- 223. ECS auto-scaling: scale out when CPU > 70%, scale in when CPU < 30%
- 224. CloudWatch alarms for all metrics defined in TAD Section 11.2

Dependencies

- [HIQ-031](#)
- [HIQ-007](#)
- [HIQ-001](#)

HIQ-060 OWASP Top 10 Audit + Security Headers + Pentest

Type: Security

Priority: P0

Points: 8

Sprint: Sprint 6 | Assignee: Security

Description

Complete the full security audit before production launch. Run OWASP ZAP scan against staging, verify all security headers present, execute all 25 penetration test cases from the SACD pentest checklist, and document all findings with resolution status.

Acceptance Criteria

- 225. OWASP ZAP scan against staging: zero High or Critical findings
- 226. All security headers present on every response (HSTS, CSP, X-Frame-Options, X-Content-Type, Referrer-Policy, Permissions-Policy)
- 227. All 25 penetration test cases from SACD Section 11 executed and documented
- 228. All REQUIRED pentest cases: PASS status
- 229. SQL injection tests: all return parameterized-safe responses
- 230. XSS injection tests: all output correctly escaped
- 231. RBAC bypass tests: all return correct 403 responses
- 232. TLS 1.0 and 1.1 confirmed disabled via SSL Labs scan (score A or A+)
- 233. Security test report committed to /docs/security-audit-v1.0.pdf

Dependencies

- [HIQ-059](#)
- [HIQ-015](#)
- [HIQ-016](#)
- [HIQ-017](#)

- HIQ-018

SUMMARY

Sprint Velocity & Completion Targets

Sprint Summary & Velocity

Points by Sprint

Sprint	Focus	Timeline	Tickets	Story Points
Sprint 1	Foundation & Setup	Weeks 1-2	10 tickets	26 points
Sprint 2	Auth & Security	Weeks 3-4	10 tickets	39 points
Sprint 3	Profiles & Jobs	Weeks 5-6	10 tickets	42 points
Sprint 4	Resume & Applications	Weeks 7-8	10 tickets	48 points
Sprint 5	AI / ML System	Weeks 9-10	10 tickets	59 points
Sprint 6	Dashboards & Deploy	Weeks 11-13	10 tickets	68 points
TOTAL	All Sprints	13 Weeks	60 tickets	282 points

Points by Type

Type	Description	Count	Points
Feature	User-facing functionality tickets	32 tickets	161 points
AI	AI/ML model and engine tickets	6 tickets	51 points
Chore	Setup, config, migrations	11 tickets	30 points
Infra	AWS, CI/CD, DevOps	5 tickets	24 points
Security	Security implementation tickets	4 tickets	19 points
Bug	Reserved for defects found in sprint	2 tickets (buffer)	—

Definition of Done

A ticket is only marked DONE when ALL of the following are true:

- Code is written, reviewed (at least 1 peer approval), and merged to main branch
- All acceptance criteria from the ticket card are verified and passing
- Unit tests written for all new business logic (minimum coverage maintained)
- No new TypeScript errors, no new ESLint violations, no new ruff violations
- CI pipeline (lint + test + build) passes on the merged commit
- Feature tested manually in the shared staging environment by the ticket author
- Any new API endpoints documented in the Swagger /docs page
- Any new environment variables added to .env.example with comments

HireIQ Feature Ticket List v1.0 — End of Document

Total: 60 tickets | 6 Sprints | 13 Weeks | All documents in the HireIQ v1.0 series are now complete.